Note 14.4 The Chain Rule

Functions of Two Variables

Theorem 5--Chain Rule for Functions of One Independent Variable and Two Intermediate Variable

It use the partial derivatives of two intermediate variable to expressed the derivative of the dependent variable to the independent variable with respect the two intermediate variable.

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

It can be obtained by the definition of differentiation.

It reflects how many times z increments when t increments and it can be understood in the aspect of composition function.

Function of Three Variables

Just need to add the third term to the two-variable formula.

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$$

Functions Defined on Surfaces

Theorem 7 Chain Rule for Two Independent Variables and Three Intermediate Variables

Suppose that w is the function with x, y, z three parameters and x, y, z have two parameters r, s respectively, then w has partial derivatives with respect to r and s.

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial r}$$

and another is similar. And w with more or less parameters are also similar.

Functions of Many Variables

Similar to upper parts.

Exercises

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